

Yumeng He

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RESEARCH INTERESTS

My research interests are physical AI, robotics, computer vision, and graphics, with a focus on object/scene generation, articulated and kinematic asset modeling, physics-based simulation, real-to-sim pipelines, and policy learning.

EDUCATION

University of California, Los Angeles

PhD of Computer Science

Sep 2026 – May 2030

Los Angeles, CA, USA

- Computer Science major
- Advisor: Prof. Chenfanfu Jiang

University of Southern California

Master of Science with Thesis

Aug 2024 – May 2026

Los Angeles, CA, USA

- Cumulative GPA: 3.91/4.0
- Computer Science major
- Advisor: Prof. Jernej Barbic

University of Toronto St. George Campus

Honours Bachelor of Science with High Distinction

Sep 2019 – May 2024

Toronto, ON, Canada

- Cumulative GPA: 3.71/4.0
- Computer Science and Mathematics double major

PUBLICATIONS AND MANUSCRIPTS

- [1] **ArticulateArena: A Metric for Articulated Kinematics**
Yumeng He, Yongfei She, Huanyu Chen, Chun Yuan, Peihao Li, Joseph Masterjohn, Yin Yang, Ying Jiang, Chenfanfu Jiang
arXiv, 2026
- [2] **One Spool at a Time: Fabricable Multicolor String Art**
Qiyang Chen, Yumeng He, Jiayin Lu, Yin Yang, Ying Jiang, Chenfanfu Jiang
SIGGRAPH Asia Poster, 2026
- [3] **Beyond Kirchhoff–Love: A Volumetric Approach to Thin-Shell Mechanics**
Huanyu Chen, Yumeng He, Jernej Barbic
ACM Transactions on Graphics (SIGGRAPH Asia), 2026
- [4] **Vibrational Affine Body Dynamics**
Haoyang Shi, Yumeng He, Sheldon Andrews, Chenfanfu Jiang, Kui Wu, Yin Yang
ACM Transactions on Graphics (SIGGRAPH Asia), 2026
- [5] **Fishbone: From One 3D Asset to a Million Controllable Edits**
Yumeng He, Xiaoying Wang, Peihao Li, Yanjia Huang, Joe Masterjohn, Jiajun Wu, Leonidas Guibas, Yin Yang, Ying Jiang, Chenfanfu Jiang
arXiv, 2026
- [6] **SeeClear: Reliable Transparent Object Depth Estimation via Generative Opacification**
Xiaoying Wang*, Yumeng He*, Jingkai Shi*, Jiayin Lu, Yin Yang, Ying Jiang, Chenfanfu Jiang
European Conference on Computer Vision (ECCV), 2026
- [7] **VoroLight: Learning Quality Volumetric Voronoi Meshes from General Inputs**
Jiayin Lu*, Ying Jiang*, Yumeng He*, Yin Yang, Chenfanfu Jiang
arXiv, 2025
- [8] **SPARK: Sim-ready Part-level Articulated Reconstruction with VLM Knowledge**
Yumeng He*, Ying Jiang*, Jiayin Lu*, Yin Yang, Chenfanfu Jiang
Computer Vision and Pattern Recognition (CVPR), 2026 **Oral (top 3.45%), CVPR Compute Transparency Champion**
- [9] **Birth of a Painting: Differentiable Brushstroke Reconstruction**
Ying Jiang*, Jiayin Lu*, Yunuo Chen*, Yumeng He, Kui Wu, Yin Yang, Chenfanfu Jiang
Conditionally Accepted by ACM Transactions on Graphics (SIGGRAPH), 2026
- [10] **Self-supervised dual-layer 2D normalizing flow method for industrial anomaly detection**
Zhenlian Miao, Guangzhu Chen, Xiaojuan Liao, Jiu Dai, Yumeng He
Applied Soft Computing (ASOC), 2024.

RESEARCH EXPERIENCE

USC Graphics Master Thesis	Aug 2025 – May 2026
<i>Supervisor: Prof. Jernej Barbic</i>	<i>Los Angeles, CA, USA</i>
Enhancing traditional Kirchhoff–Love shell models by introducing a novel elastic energy formulation that accurately captures stretching, compression, and bending, while overcoming KL shells’ inability to reproduce Poisson effects	
UCLA AIVC Lab Visiting Student	Jun 2025 – Aug 2026
<i>Supervisor: Prof. Chenfanfu Jiang</i>	<i>Los Angeles, CA, USA</i>
Leading research on end-to-end articulation-aware 3D mesh generation, designing a pipeline that synthesizes multi-part objects from text or image conditions while mitigating over-segmentation via per-part image-guided local attention	
USC RESL Lab Research Assistant	Jul 2025 – present
<i>Supervisor: Prof. Gaurav S. Sukhatme</i>	<i>Los Angeles, CA, USA</i>
Designed a few-shot robotic manipulation task that uses Model Predictive Control to optimize material parameters for physically accurate simulation	
CDUT Research Assistant	May 2023 - Aug 2023
<i>Supervisor: Prof. Guangzhu Chen</i>	<i>Remote</i>
Faced with unlabeled industrial defects on MVTec AD and tasked to raise unsupervised detection/localization accuracy, proposed SS-DualFlow: a dual-layer 2D normalizing-flow that maps features to a Gaussian base to curb information loss and inserts an Exponential Space Attention module to focus on anomaly-salient regions. Result: image-level AUROC = 99.38% and pixel-level AUROC = 98.38% on MVTec AD; co-authored the Applied Soft Computing (2024) paper.	

RELEVANT PROJECTS

Incompressible Fluid Simulation: A Comparison C++, OpenGL	Spring 2025
Implemented and benchmarked four 2D incompressible fluid solvers, including Stable Fluids, Smoothed Particle Hydrodynamics (SPH), Particle In Cell (PIC), and Affine Particle In Cell (APIC) on identical scenarios, reporting performance-vs-accuracy trade-offs.	
Motion Capture Interpolation C++	Spring 2025
Implemented four interpolation schemes (including linear & Bézier) with both Euler- and quaternion-based rotations to reconstruct and smooth optical mocap, mitigating gimbal lock and interpolation artifacts	

INDUSTRY EXPERIENCE

Software Developer Intern Full Time	Aug 2022 – Aug 2023
<i>HCL Canada Inc.</i>	<i>Toronto, ON, Canada</i>
Automated build and deployment pipelines using Jenkins, accelerating production releases and minimizing manual errors by eliminating key bottlenecks in CI/CD	

COMMUNITY SERVICES

Reviewer: SIGGRAPH Asia, Pacific Graphics, TVCG	
Member of Society of Women Engineers (SWE)	2025 - 2026
Mentor for Viterbi Graduate Mentorship Program	Fall 2025
Mentor for Women in Engineering Mentorship	Fall 2025
Mentor for Viterbi Graduate Mentorship Program	Summer 2025

INVITED TALKS

<i>MPC Is All You Need.</i> [Co-presented]. USC SLURM Lab, hosted by Prof. Daniel Seita	Aug 2025
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AWARDS & HONORS

Dean’s List Scholar	2020 - 2024
Ranked in the top 25% of contestants in the Galois Contest	2017
Ranked in the top 25% of contestants in the Cayley Contest	2017
Ranked in the top 25% of contestants in the Canadian Intermediate Mathematics	2016

TECHNICAL SKILLS

Programming Tools: Linux, Windows, MacOS, C++, Python, HTML, CSS, JavaScript, MySQL, R, OpenGL, GLSL, Eigen, Docker, Jenkins, Gradle, Ant, Shell, Git, Github Actions	
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